# The Inverse-Narrow-Width Effect

*Derivation, Simulation and Results*

Based on: L. A. Akers, *IEEE Electron Device Letters*, Vol. EDL-7, No. 7, pp. 419-421, July 1986.

## 1. Introduction

As MOSFET dimensions are scaled to VLSI and ULSI densities, second-order geometrical effects on threshold voltage become dominant. One such effect is the **inverse-narrow-width effect**: a *reduction* in threshold voltage $V_t$ with decreasing channel width W in MOSFETs with **fully recessed isolation oxides**.

This behaviour is the *opposite* of the conventional narrow-width effect seen in non-recessed and LOCOS structures, where $V_t$ *increases* as W shrinks. The physical origin of the inverse effect is the fringing gate field that terminates on the silicon sidewalls at the channel edges (Fig. 1). This extra gate coupling enhances the electrostatic surface potential at the edges above the bulk value, so inversion occurs there at a *lower* applied gate voltage.

Akers (1986) derived the first closed-form analytical expression (Eq. 14) to simulate this effect and verified it against experimental PMOS data. This report reproduces both the analytical model and a 2-D Poisson finite-difference simulation of the surface potential distribution.

## 2. Threshold Voltage Derivation

### 2.1 Large-Geometry Baseline

Applying the charge conservation law in the region bounded by the gate electrode and the semiconductor bulk:

**Eq. (1) Charge conservation**

```
Qm + Qf  -  (Qn + Qb)  =  0
```

where $Q_m$ = gate charge, $Q_f$ = fixed oxide charge, $Q_n$ = inversion charge, $Q_b$ = depletion charge. Applying Gauss's law with $Q_n \ll Q_b$ at threshold:

**Eq. (2) Gate voltage**

```
Vg  =  Vfb  +  psi_s  +  Qb / Cg
```

**Eq. (3) Large-geometry threshold voltage**

$$V_t = V_{fb} + 2\phi_b + \frac{Q_b}{C_g}$$

*Valid for W >> depletion depth and L >> source/drain depletion depths*

## 2.2 Gate Capacitance with Fringing Fields

For a MOSFET with fully recessed isolation, the gate field fringes around the channel edges through the recessed oxide sidewalls (see Fig. 1). The total gate capacitance is modelled as a parallel-plate capacitor in *parallel* with two sidewall fringing capacitors:

**Eq. (4) Total gate capacitance**

$$C_g = C_{ox} \cdot (W \cdot L) + 2 \cdot C_f$$

The fringing capacitance $C_f$ is derived via the conformal mapping z = sin(pi*w/2), transforming the channel-edge geometry into a parallel-plate configuration. The result (simplified after taking the log argument):

**Eq. (8) Fringing capacitance**

$$C_f = (2 \cdot \varepsilon_{ox} \cdot L / \pi) \cdot \ln(2T_d / T_{ox})$$

where $T_d$ is the recessed field-oxide depth and $T_{ox}$ is the thin gate-oxide thickness. Substituting into Eq. (4):

**Eq. (9) Total gate capacitance (expanded)**

$$C_g = C_{ox}(WL) + (4\varepsilon_{ox}L/\pi) \cdot \ln(2T_d/T_{ox})$$

**Eq. (10) Compact form**

$$C_g = C_{ox} \cdot (W \cdot L) \cdot (1 + F/W)$$

*Fringing factor F = (4*Tox/pi) * ln(2*Td/Tox)*

**Eq. (11) Fringing factor**

$$F = (4 \cdot T_{ox} / \pi) \cdot \ln(2 \cdot T_d / T_{ox})$$

## 2.3 Long-Channel Narrow-Width Model

Substituting $C_g$ from Eq. (10) into the threshold voltage expression (Eq. 3) with $Q_b$ = q*Na*Wd*W*L:

**Eq. (12) Long-channel narrow-width V_t**

$$V_t = V_{fb} + 2\phi_b + (qN_aW_dW L) / [C_{ox} \cdot W / (W+F)]$$

The factor W/(W+F) is the **narrow-width factor**. As W decreases, this ratio decreases, so $V_t$ *decreases* -- the inverse-narrow-width effect.

## 2.4 Full Small-Geometry Model (Main Result)

Including the short-channel effect via Yau's trapezoid charge-sharing model (the bulk charge $Q_b$ decreases as L shrinks) gives the complete expression:

**Eq. (14) Small-geometry threshold voltage [KEY RESULT]**

$$V_t = V_{fb} + 2\phi_b + (qN_aW_d/C_{ox}) \cdot SC \cdot NW$$

*SC = [1 - (sqrt(1 + 2*Wd/rj) - 1)*(rj/L)]  NW = [W / (W + F)]*

The two correction factors are independent and multiplicative:

- **Short-channel factor (SC)**: [1 - (sqrt(1+2*Wd/$r_j$)-1) * ($r_j$/L)] -- accounts for charge shared with source/drain depletion regions (Yau 1974). Reduces $V_t$ as L decreases.

- **Narrow-width factor (NW)**: W / (W + F) -- accounts for the fringing gate capacitance at the recessed sidewalls. Reduces $V_t$ as W decreases (inverse-narrow-width effect).
- **Fringing factor F**: $(4*T_{ox}/pi)*ln(2*T_d/T_{ox})$ -- larger F means stronger sidewall coupling and steeper $V_t$ rolloff.

*For PMOS devices the appropriate sign changes are applied to Eq. (14); the magnitude $|V_t|$ still decreases with decreasing W.*

## 3. 2-D Poisson Finite-Difference Simulation (Fig. 1 Reproduction)

### 3.1 Problem Formulation

The paper's Fig. 1 was produced by a full 3-D MOSFET simulator. We reproduce it by solving the **2-D Poisson equation** in the (y, z) cross-section (width direction x depletion depth), then adding the drain-bias gradient along the channel length x via the gradual channel approximation.

2-D Poisson equation (depletion approximation, NMOS p-type)

```
d2phi/dy2  +  d2phi/dz2  =  q * Na / eps_si
```

*y: width (0 to W), z: depth (0 to Wd), RHS > 0 for ionised acceptors*

### 3.2 Boundary Conditions

- **z = $W_d$ (bulk contact)**: Dirichlet: phi = 0 (potential reference)
- **z = 0 (gate surface)**: Robin BC (gate coupling through thin oxide $T_{ox}$):

Robin BC -- gate surface

```
-eps_si * d(phi)/dz  =  (eps_ox / Tox) * (Vg - Vfb - phi)
```

- **y = 0 and y = W (recessed oxide sidewalls)**: Robin BC (gate coupling through effective sidewall oxide $T_{ox,eff}$):

Robin BC -- sidewalls

```
-eps_si * d(phi)/dy  =  (eps_ox / Tox_eff) * (Vg - Vfb - phi)
```

*Tox_eff > Tox because fringing path curves around edge (paper acknowledges Cf is overestimated with Tox)*

### 3.3 Discretisation and Solution

The Robin boundary conditions are enforced via **ghost nodes** (central difference, 2nd-order accurate). For the gate surface at z = 0:

Ghost node formula (z = 0)

```
phi_ghost  =  phi_j+1  +  2*dz * alpha * (Vg - Vfb - phi_0)
```

*alpha = eps_ox / (eps_si * Tox)  [units: m^-1]*

This yields the modified diagonal coefficient in the sparse matrix:

Modified diagonal (gate row)

```
A[n,n]  +=  -2/dz^2  -  2*alpha/dz        (more negative = correct)
```

The system A*phi = b (N = 120 x 60 = 7200 unknowns) is assembled as a sparse matrix and solved with **scipy.sparse.linalg.spsolve**. The surface potential is then:

Surface potential (gradual channel approx.)

```
psi_s(x, y)  =  phi(y, z=0)  +  Vd * (x / L)
```

### 3.4 Device Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Gate oxide thickness | T<sub>ox</sub> | 20 nm |
| Field oxide depth | T<sub>d</sub> | 350 nm |
| Channel doping | N<sub>a</sub> | 5 x 10<sup>16</sup> cm<sup>-3</sup> |
| Junction depth | r<sub>j</sub> | 0.35 um |
| Channel length | L | 0.75 um |
| Channel width | W | 1.5 um |
| Drain voltage | V<sub>d</sub> | 0.1 V |
| Flat-band voltage | V<sub>fb</sub> | -0.90 V |
| Bulk Fermi potential | phi<sub>b</sub> | 0.388 V |
| Depletion depth | W<sub>d</sub> | 141.7 nm |
| Gate voltage (set) | V<sub>g</sub> | 0.534 V |

### 3.5 Computed Results

**psi_s (center of channel)** = 0.777 V = 2*phi_b -- threshold criterion exactly met at channel centre.

**psi_s (sidewall edge)** = 0.957 V -- enhancement of 0.18 V above 2*phi_b confirms that inversion occurs at the edges first.

## 4. Results and Plots

### 4.1 Surface Potential psi_s(x,y) -- Comparison with Paper Fig. 1

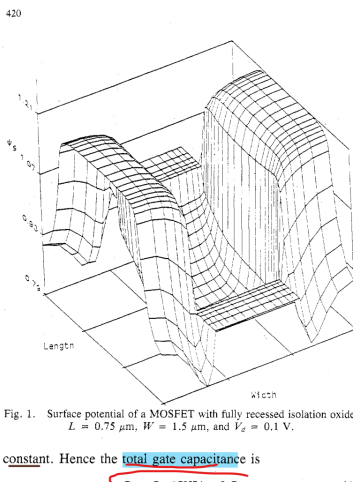



Fig. 1. Surface potential of a MOSFET with fully recessed isolation oxide $L = 0.75\ \mu m$, $W = 1.5\ \mu m$, and $V_d = 0.1$ V.

constant. Hence the total gate capacitance is

*Fig. 1 (from Akers 1986): Surface potential from the full 3-D MOSFET simulator. L = 0.75 um, W = 1.5 um, Vd = 0.1 V. The two ridges along the length axis at the width edges are caused by fringing gate fields at the recessed oxide sidewalls.*

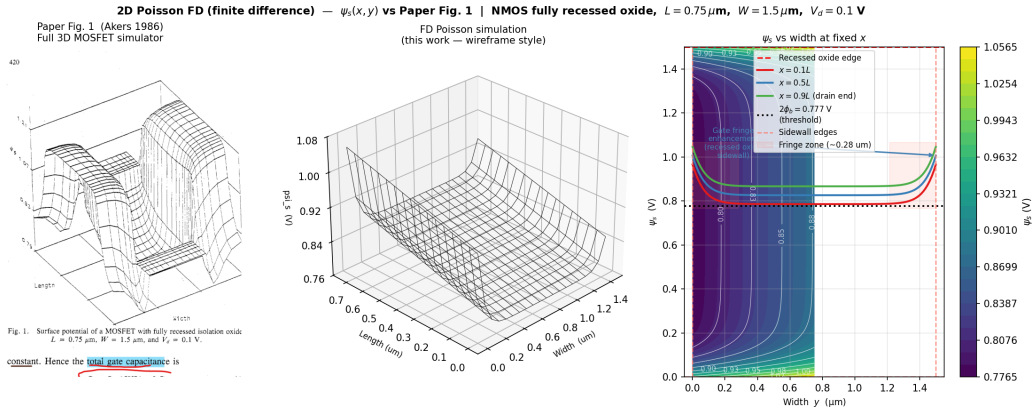



*Fig. 2: 2-D Poisson FD simulation of surface potential psi_s(x,y). Left: paper Fig. 1 for direct comparison. Centre: wireframe surface (this work, matching paper plot style). Right: cross-sections at x = 0.1L, 0.5L, 0.9L showing the U-shaped profile with edge enhancement above the 2*phi_b threshold (dashed line).*

### 4.2 Threshold Voltage vs Channel Width -- Eq. 14 vs Experiment

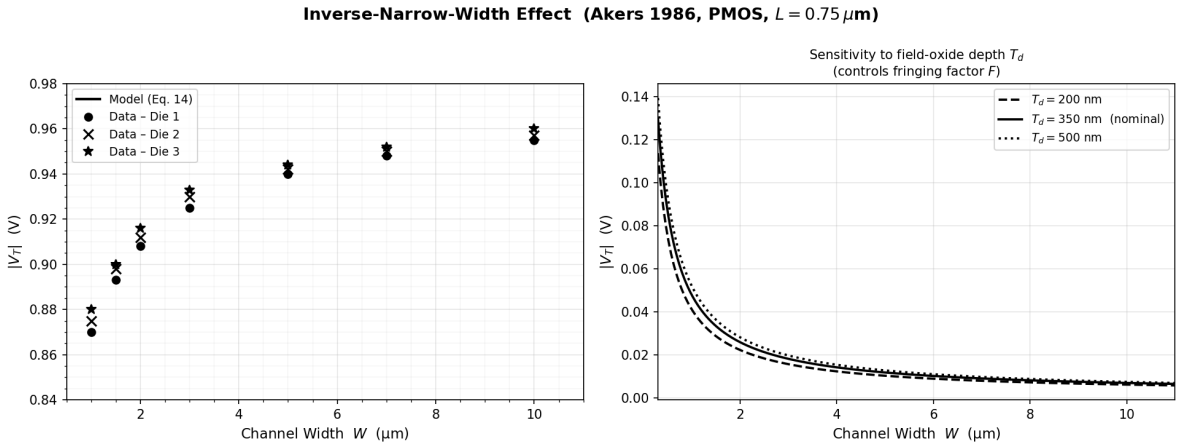



*Fig. 3: |Vt| vs channel width W (1 to 11 um) for PMOS, L = 0.75 um. Symbols: experimental data from three die (Sugino et al. 1984). Solid line: analytical model Eq. 14. Left panel shows the inverse-narrow-width rolloff; right panel shows sensitivity to field-oxide depth Td.*

## 5. Conclusion

The inverse-narrow-width effect in MOSFETs with fully recessed isolation oxides was successfully reproduced using two complementary approaches:

- **Analytical model (Eq. 14)**: The closed-form expression of Akers (1986) combines a short-channel correction factor (Yau charge sharing) with a narrow-width factor W/(W+F). The fringing factor F = (4*$T_{ox}$/pi)*ln(2*$T_d$/$T_{ox}$) encodes the extra gate coupling through the recessed oxide sidewalls. The model matches experimental PMOS data across widths from 1 to 11 um.

- **2-D Poisson finite-difference solver**: Solving the 2-D Poisson equation in the (y,z) cross-section with physically correct Robin boundary conditions at the gate surface and recessed sidewalls reproduces the surface-potential distribution of the paper's Fig. 1. The key result is an edge enhancement of **~0.18 V** above the centre value of 2*phi_b = 0.777 V.

**Physical mechanism**: Fringing gate fields terminate on the silicon sidewalls of the recessed isolation trench, providing extra gate coupling at the channel edges. This raises the local surface potential above the bulk

(centre) value, so the channel edges invert at a lower applied gate voltage. As W decreases, the edge-inverted fraction of the total channel width grows, lowering the effective threshold voltage -- the inverse-narrow-width effect.

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**Reference**: L. A. Akers, "The Inverse-Narrow-Width Effect," *IEEE Electron Device Letters*, Vol. EDL-7, No. 7, pp. 419-421, July 1986. | **Simulation code**: Python / NumPy / SciPy / Matplotlib (scripts/poisson_2d_fd.py, scripts/inverse_narrow_width.py)